NETWORKING

TCP Teardown & Reliability

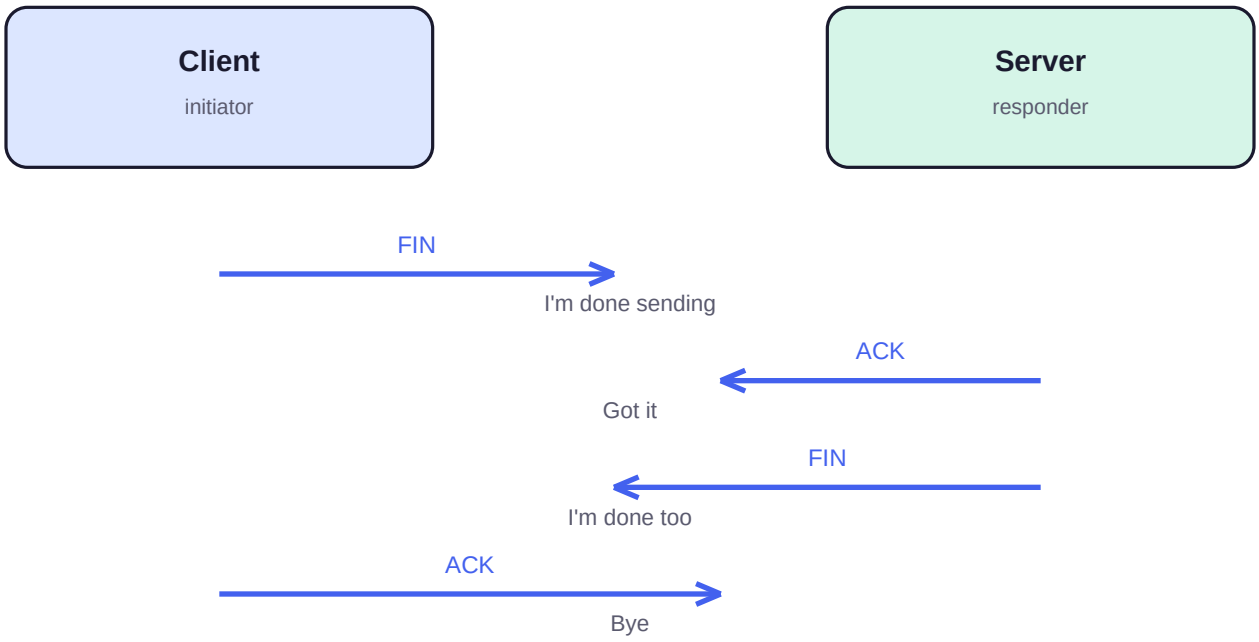
Closing connections cleanly —
and the machinery that makes TCP trustworthy.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

The FIN/ACK Close Sequence

A graceful goodbye is a 4-way handshake.



Each side closes its half-duplex direction independently — a connection is two one-way streams.

Simultaneous close: both sides send FIN at once — the sequence still converges, with a slightly different ACK pattern.

STEP 2/5

TIME_WAIT: Why the Wait?

The side that sends the final ACK lingers for 2×MSL (~60s).

TIME_WAIT state

socket held 2×MSL after final ACK — then fully closed

- Reason 1 — lost final ACK: if the last ACK is dropped, the peer retransmits FIN; without TIME_WAIT nobody would answer, and the peer would RST.
- Reason 2 — delayed segments: old duplicate packets from this connection could corrupt a NEW connection reusing the same ports; the wait lets them die.
- Cost: ports held in TIME_WAIT can't be reused — busy servers tune this (SO_REUSEADDR, shorter MSL) and watch for port exhaustion.

STEP 3/5

Retransmission: Timers & Fast Recovery

No ACK in time? Send it again — but smartly.

RTO — Retransmission Timeout

per-segment timer; on expiry, resend and back off exponentially

Fast retransmit

3 duplicate ACKs → resend NOW, don't wait for the timer

SACK — Selective ACK

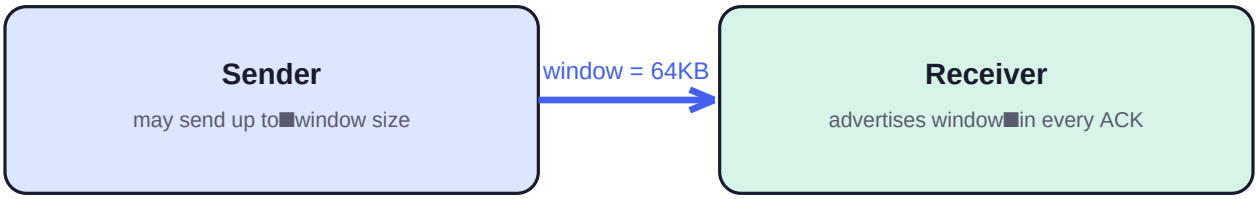
receiver says exactly which blocks arrived; resend only the gaps

RTO is computed from measured RTT (smoothed) — too short wastes bandwidth, too long stalls the connection.

STEP 4/5

Sliding Window: Flow Control

The receiver throttles the sender — never overflow a slow reader.



- Window = free buffer space at the receiver; advertised in every ACK header.
- Window slides forward as data is ACKed — the sender keeps the pipe full without drowning the receiver.
- Zero window = 'stop!' — sender probes periodically until the window reopens. A stuck zero window = stalled connection.

STEP 5/5

Congestion Control: CUBIC

Flow control protects the receiver; congestion control protects the NETWORK.

Slow start

cwnd doubles per RTT — exponential probe for capacity

Loss event

packet loss = congestion signal → cut cwnd sharply

CUBIC growth

window grows as a cubic function of time since last loss — fast recovery, gentle near the ceiling

CUBIC is the Linux/macOS default: it finds high-bandwidth paths quickly and fairly shares with other flows.

BBR is the modern alternative: models the bottleneck's bandwidth and RTT instead of reacting to loss — better on lossy links.

STEP ✓/✓

Key Takeaways

TCP Teardown & Reliability — Remember

- Graceful close is a 4-way FIN/ACK exchange — two independent half-closes.
- TIME_WAIT (2×MSL) guards against lost final ACKs and zombie segments.
- Retransmission: RTO timers, fast retransmit on 3 dup-ACKs, SACK for precision.
- Sliding window = receiver-driven flow control; zero window means stop.
- Congestion control (CUBIC) protects the network: slow start, then cubic growth after loss.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev